

DYNAMIC CONTEXTUAL PRICING ENGINE WITH MULTI-ARMED BANDITS (THOMPSON SAMPLING)

ABSTRACT:

This project presents a dynamic pricing system that uses the Thompson Sampling algorithm to determine the best product price. Different price levels are treated as bandit arms, and the system learns from customer purchase behavior. By continuously updating its decisions, the model aims to maximize revenue while adapting to changing market conditions.

INTRODUCTION:

Dynamic pricing is a strategy where product prices change based on customer demand, competition, and market trends. Traditional pricing methods are fixed and may not maximize profits. Thompson Sampling, a Multi-Armed Bandit algorithm, helps businesses find the best price by balancing exploration of new prices and exploitation of successful ones.

THEORY:

The Multi-Armed Bandit model considers each price as an arm. Thompson Sampling uses Bayesian probability with Beta distributions to estimate the success rate of each price. After every customer interaction, the probability is updated, allowing the system to improve pricing decisions over time.

OBJECTIVES:

- Develop a dynamic pricing engine using Thompson Sampling.
- Select the most profitable price based on customer responses.
- Improve revenue and conversion rates.
- Adapt to changing market conditions.

METHODOLOGY:

1. Define multiple product price levels.
2. Initialize Beta distributions for each price.
3. Select a price using Thompson Sampling.
4. Record the customer's purchase response.
5. Update the Beta distribution based on the result.
6. Repeat the process to improve future pricing decisions.

DATASET:

The dataset includes customer and product information such as:

- Product Price
- Customer ID
- Purchase Status (Yes/No)
- Competitor Price
- Time of Purchase
- Revenue

THOMPSON SAMPLING MECHANICS:

- Initialize each price with a Beta distribution.
- Randomly sample from each distribution.
- Choose the price with the highest sampled value.
- Update the distribution after observing whether the customer purchased the product.

COMPLEXITY ANALYSIS:

- Price Selection: $O(k)$, where k is the number of price options.
- Distribution Update: $O(1)$
- Overall Complexity: $O(k)$ per iteration.

RESULTS:

The proposed system gradually learns the most effective pricing strategy. Compared to fixed pricing, it improves revenue and customer conversion by selecting prices based on real-time customer behavior.

APPLICATIONS:

- E-commerce websites
- Retail stores
- Airline ticket pricing
- Hotel room booking
- Ride-sharing services

CONCLUSION:

The Dynamic Contextual Pricing Engine using Thompson Sampling provides an efficient solution for real-time price optimization. It continuously learns from customer behavior, adapts to market changes, and helps businesses maximize revenue through intelligent pricing decisions.